

Qn	Solution
1(a)	$[x] = m$ $[E] = \frac{[F]}{[A]} = \frac{\text{kg m s}^{-2}}{\text{m}^2} = \text{kg m}^{-1} \text{s}^{-2}$ $[g] = [a] = \text{m s}^{-2}$ $\left[\frac{L^3 g M}{wt^3 E} \right] = \frac{(\text{m}^3)(\text{m s}^{-2})(\text{kg})}{(\text{m})(\text{m}^3)(\text{kg m}^{-1} \text{s}^{-2})} = m$ Since the terms on both sides of the equation has the same SI base unit, the equation is homogenous.
1(b)	Using $x = \frac{L^3 g M}{4wt^3 E}$, $E = \frac{L^3 g M}{4wt^3 x}$ $= \frac{(0.800)^3 (9.81)(0.300)}{4(2.10 \times 10^{-2})(4.56 \times 10^{-3})^3 (1.00 \times 10^{-2})}$ $= 1.8918 \times 10^{10}$ $\approx 1.89 \times 10^{10} \text{ Pa}$ $\frac{\Delta E}{E} = 3 \frac{\Delta L}{L} + \frac{\Delta M}{M} + \frac{\Delta w}{w} + 3 \frac{\Delta t}{t} + \frac{\Delta x}{x} \quad (g \text{ is a constant})$ $= 3 \left(\frac{0.005}{0.800} \right) + \left(\frac{2}{300} \right) + \left(\frac{0.02}{2.10} \right) + 3 \left(\frac{0.01}{4.56} \right) + \left(\frac{0.01}{1.00} \right)$ $= 0.0515$ ≈ 0.05 $= 5\%$ $\therefore E = 1.89 \times 10^{10} \text{ Pa} \pm 5\%$
2(a)(i)	